TradeLearn – Learn skills. Build real values

PROJECT DESCRIPTION DOCUMENT

Submitted for Copyright Registration

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1. INTRODUCTION

TradeLearn is a web-based peer-to-peer skill exchange platform designed to create a collaborative learning environment where users can teach and learn skills without relying on monetary transactions. Traditional online learning systems often depend on paid subscriptions and one-way content delivery methods, limiting accessibility and active user participation. TradeLearn addresses this issue by introducing a credit-based model where knowledge itself acts as a valuable resource.

In today's rapidly evolving digital environment, individuals continuously need to acquire new skills to improve academic performance, professional growth, and career opportunities. However, many learners face financial barriers that prevent access to premium educational resources. Simultaneously, many people already possess valuable skills and expertise but lack an organized platform where they can exchange knowledge for mutual benefit.

TradeLearn creates an ecosystem where users become both learners and contributors. Through a structured exchange process, users can teach others, earn credits, and use those credits to learn new skills. The system combines profile management, request handling, authentication, and skill discovery into a unified learning platform.

2. OBJECTIVE

The primary objective of TradeLearn is to develop a web-based platform that enables users to exchange skills through a structured peer-to-peer learning mechanism. The system aims to eliminate financial barriers by introducing a credit-based exchange model where users can earn credits through teaching and utilize them for learning.

Another objective is to promote collaborative learning by encouraging active participation rather than passive content consumption. The project also focuses on implementing secure authentication, request management, credit tracking, and efficient skill discovery mechanisms while applying practical concepts of full-stack web development.

3. KEY FEATURES

- **User Registration and Authentication**
  Secure account registration and login functionality using authentication mechanisms to provide protected access and maintain user privacy.

- **Profile Management**
  Users can create and update personalized profiles by adding skills they possess and skills they wish to learn.

- **Skill Exchange Requests**
  Users can send, accept, reject, and manage learning requests for desired skills.

- **Credit-Based Learning System**
  Users earn credits by teaching others and spend credits to acquire new skills.

- **Dashboard Management**
  Provides users with a centralized dashboard displaying credits, request history, exchange progress, and ongoing activities.

- **Request Tracking System**
  Tracks request status such as pending, accepted, rejected, and completed.

- **Search and Skill Discovery**
  Allows users to search and browse available skills and discover suitable learning partners

4. TECHNOLOGY STACK

TradeLearn is developed using a full-stack web development architecture that combines frontend technologies, backend frameworks, database systems, authentication mechanisms, and development tools. frontend is built using HTML, CSS, and EJS, enabling dynamic content rendering and responsive user interfaces. The backend uses Node.js and Express.js to manage business logic, route handling, middleware integration, and API processing.
MongoDB serves as the primary database for storing user profiles, skill information, requests, credits, and activity data. Authentication and authorization are handled using JWT (JSON Web Token) to ensure secure user access and session management.
Development tools such as Git, GitHub, and Postman are used for version control, collaboration, and API testing.

5. SYSTEM ARCHITECTURE

TradeLearn follows a full-stack layered architecture where the frontend communicates with backend services through APIs. The backend processes business logic, request handling, authentication, and credit calculations, while MongoDB manages data storage.
Users create profiles, add offered skills and desired skills, search for learning opportunities, and initiate exchange requests. The system processes user interactions, updates credit balances, and maintains records of learning activities. The modular design improves scalability, maintainability, and performance.

6. SECURITY AND DATA HANDLING

Security is a major aspect of TradeLearn. Authentication mechanisms ensure only authorized users can access the platform.
The system implements **JWT-based authentication**, secure password handling, middleware validation, and protected routes to maintain privacy and prevent unauthorized access.
User information such as profiles, credits, and request data is securely stored within the database, ensuring data integrity and protection.

7. FUNCTIONAL OVERVIEW

TradeLearn provides a centralized environment where users can manage profiles, skill offerings, learning requests, and credit transactions.
The system allows users to participate in peer-to-peer learning activities through a structured workflow. Users browse skills, connect with other members, send requests, participate in learning activities, and track progress through dashboards.
The credit-based mechanism ensures active participation and rewards users who contribute knowledge to the community.

8. ORIGINALITY STATEMENT

TradeLearn represents an original implementation designed specifically to solve limitations in traditional online learning systems. Unlike conventional educational platforms that depend on monetary transactions, TradeLearn introduces a knowledge-as-currency approach where users exchange skills through credits.

The system workflow, database structure, credit mechanism, request lifecycle, and integrated learning process are independently designed and implemented. The project combines educational objectives with practical full-stack development concepts to create a unique and collaborative learning ecosystem.

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